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EDUCATION

Arab Academy for Science, Technology and Maritime Transport (AASTMT) — Alexandria, Egypt
B.Sc. in Computer Engineering — Expected May 2028
GPA: 3.4 / 4.0
Relevant Coursework: Computer Architecture, Systems Programming, Data Structures and Algorithms, DSP, Object-Oriented Programming, Embedded Systems, Data Acquisition Systems, Computer Networks.

TECHNICAL SKILLS

Languages: C · C++ (C++17/20) · Embedded C · CUDA C++ · Python · Java
GPU & Parallel Computing: CUDA · cuBLAS · TF32 Tensor Cores · kernel optimization · memory coalescing · SIMD
Deep Learning & Computer Vision: PyTorch · TensorFlow · OpenCV · YOLO · NumPy · Pandas · Matplotlib
Embedded & Robotics: Raspberry Pi · OAK-D S2 · NVIDIA Jetson · Arduino GIGA R1 · DepthAI · IMU sensor fusion · Error-State Kalman Filter · PID control · Visual-Inertial SLAM · real-time systems
Tools: Linux · bash · Git · CMake · Docker · Numba
Systems: Operating systems · Memory & resource management · scheduling · Real-time inference

EXPERIENCE

Computer Vision & Autonomy Lead, MATE ROV Team — Sep 2025 – Present
AASTMT (AAST-Triton ROVs) — Alexandria, Egypt

- Lead computer vision and autonomy on a 40-member team that took **1st place regionally** at the MATE ROV competition and **3rd place at the World Championship**.
- Designed OS-VIO, a visual-inertial SLAM pipeline for underwater navigation. It fuses stereo vision with a 15-DOF error-state Kalman filter and uses manifold pre-integration for real-time state estimates.
- Moved feature tracking and RGB/depth sync onto the OAK-D S2's VPU with DepthAI, which keeps five camera streams running at **20 FPS** on a Raspberry Pi 4.
- Migrating the ROV perception stack from Raspberry Pi 4 to an NVIDIA Jetson platform; evaluating JetPack, CUDA, and TensorRT for the underwater CV pipeline.
- Rewrote the visual Jacobians with Numba JIT (**5–10× faster on ARM**) and switched the Kalman solver to Cholesky (**2× faster** than a general inverse). Added ZUPT and pinned threads to fixed CPU cores to cut scheduler jitter.
- Built a stereo measurement tool with motion-gated depth buffering, percentile filtering that adapts to water turbidity, and refraction correction. Pool dimensions came out within **5 cm**.
- Contributed to a PID flight controller for the Arduino GIGA R1 that gave centimeter-level positioning during competition tasks.

PROJECTS

Vecore: GPU-Accelerated Deep Learning Framework — C++17, CUDA, cuBLAS
github.com/Marwan7042/Vecore

- Built a PyTorch-style framework from scratch in C++17 and CUDA with reverse-mode autodiff, N-dimensional tensors, an SGD optimizer, and a custom STL.
- Wrote CUDA kernels by hand (fused ReLU, SGD, cross entropy, broadcasting) and integrated cuBLAS with **TF32 Tensor Cores**; added an async double-buffered training pipeline and a simple caching allocator that avoids repeated cudaMalloc calls.
- Reached **96.23% MNIST test accuracy**, surpassing an equivalent PyTorch baseline (95.61%).

VCLite: Zero-Allocation Inference Engine for Edge & Space — C++17, OpenMP SIMD
github.com/Marwan7042/VCLite

- Wrote a CNN inference engine for satellites and microcontrollers that never allocates dynamically; all memory comes from one pre-allocated static pool.
- Implemented Conv2d, MaxPool2d and Dense layers with SIMD, plus Schraudolph's fast exp approximation to save power.
- Tested in Docker under a **50 MB RAM / 5% CPU** limit; it runs sliding-window inference on 4K satellite images.

libvc: Custom C++ Standard Library — C++17
github.com/Marwan7042/libvc

- Implemented a vector, matrix, hash map, circular-buffer queue and linked lists, following the Rule of Five with a zero-leak implementation.
- Stored matrices as a flat 1D array for cache locality, built the stack on top of the vector, and wrote quicksort, insertion sort and binary search.

AWARDS

- **1st Place**, MATE ROV Regional Championship
- **3rd Place**, MATE ROV International World Championship